P-values

A p-value (probability value) measures the strength of evidence against the null hypothesis H_0.

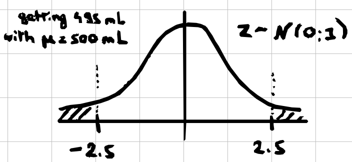
- The stronger the p-value, the stronger the evidence
  If p-value $< \alpha$ (ex: 0.05), reject H_0, if p-value $> \alpha$ we fail to reject.

Two-sided Tests

Soda bottling machine needs to fill exactly 500 mL.
So $H_0: \mu = 500$ mL (machine works perfectly)

After collecting data (36 bottles) we have $\bar{x} = 496$ mL, $\sigma = 12$ mL.
Now we calculate z-test statistic:

$$Z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{495 - 500}{12 / \sqrt{36}} = -2.5$$ (mean sits 2.5 std. error below)



We mapped our data to normal distribution.

INTUITION
If the factory was truly filling exactly at 500 mL how rare would it be to get 495 mL or less / 505 mL or more?

P-value = $P(z \leq 2.5) + P(z \geq 2.5) = 1.24\%$
H_0 is rejected, p-value $< \alpha$ (5%), machine must be miscalibrated.

One Sided Tests

Now $H_0: \mu \geq 500$ mL (bottles contain at least advertised amount)
So p-value = $P(Z \leq -2.5) = 0.62\%$ and we reject H_0.